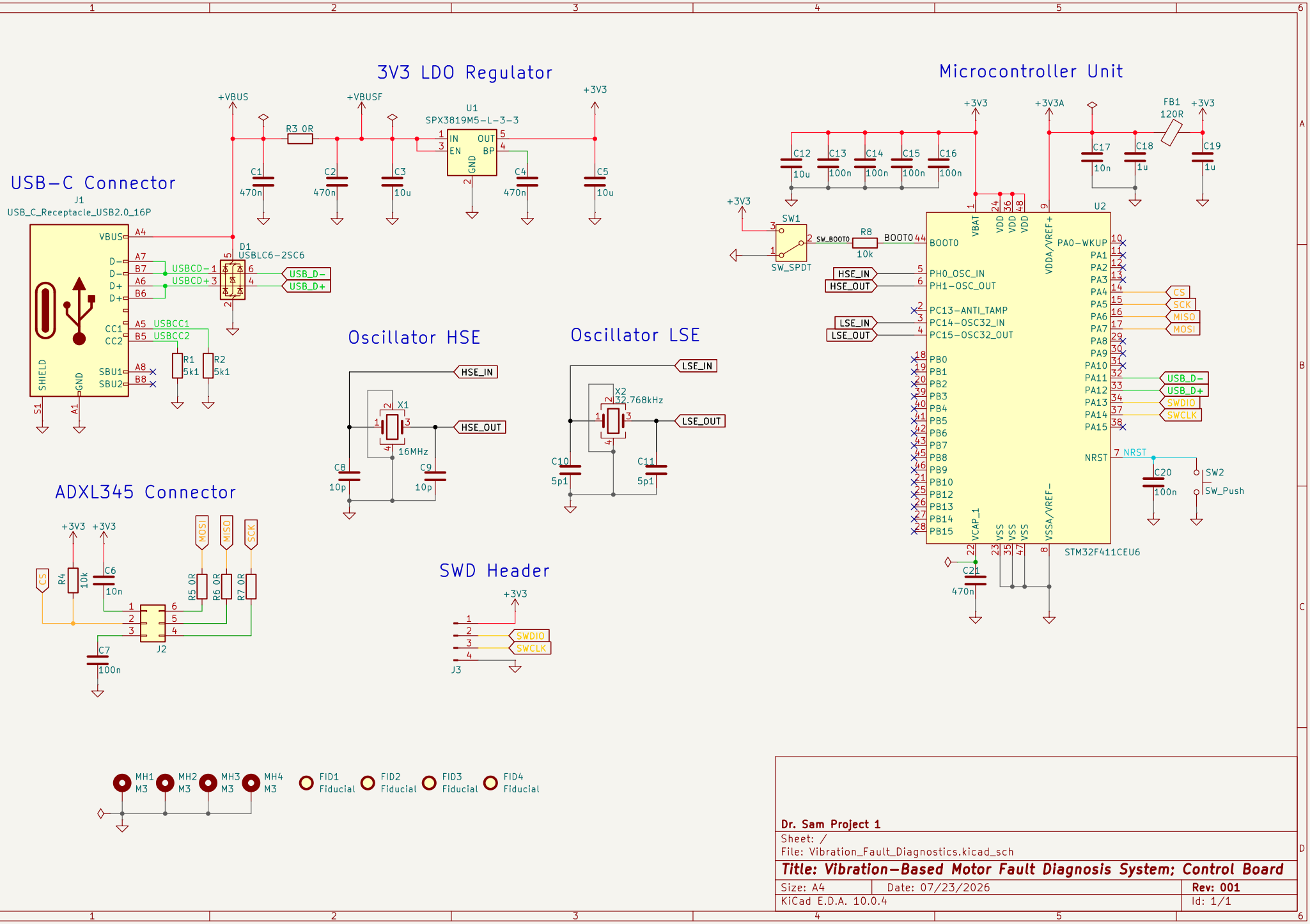

3V3 LDO Regulator
+VBUS
+VBUSF
+3V3
U1
SPX3819M5-L-3-3
IN
EN
OUT
BP
GND
R3 0R
C1
470n
C2
470n
C3
10u
C4
470n
C5
10u
USB-C Connector
J1
USB_C_Receptacle_USB2.0_16P
VBUS
D-
D+
CC1
CC2
SHIELD
GND
SBU1
SBU2
D1
USBLC6-2SC6
USBCD-
USBCD+
USB_D-
USB_D+
USBCC1
USBCC2
R1
5k1
R2
5k1
S1
A1
Oscillator HSE
HSE_IN
HSE_OUT
X1
16MHz
C8
10p
C9
10p
Oscillator LSE
LSE_IN
LSE_OUT
X2
32.768kHz
C10
5p1
C11
5p1
ADXL345 Connector
+3V3
CS
R4
10k
C6
10n
MOSI
MISO
SCK
R5 0R
R6 0R
R7 0R
J2
C7
100n
SWD Header
SWDIO
SWCLK
J3
Microcontroller Unit
+3V3
+3V3A
FB1
120R
C12
10u
C13
100n
C14
100n
C15
100n
C16
100n
C17
10n
C18
1u
C19
1u
SW1
SW_SPDT
SW_BOOT0
R8
10k
BOOT0
HSE_IN
HSE_OUT
LSE_IN
LSE_OUT
U2
VBAT
VDD
VDDA/VREF+
PA0-WKUP
PC13-ANTI_TAMP
PC14-OSC32_IN
PC15-OSC32_OUT
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
NRST
VCAP_1
VSS
VSSA/VREF-
STM32F411CEU6
CS
SCK
MISO
MOSI
USB_D-
USB_D+
SWDIO
SWCLK
C20
100n
SW2
SW_Push
C21
470n
MH1
M3
MH2
M3
MH3
M3
MH4
M3
FID1
Fiducial
FID2
Fiducial
FID3
Fiducial
FID4
Fiducial
Dr. Sam Project 1
Sheet: /
File: Vibration_Fault_Diagnostics.kicad_sch
Title: Vibration-Based Motor Fault Diagnosis System; Control Board
Size: A4
Date: 07/23/2026
Rev: 001
KiCad E.D.A. 10.0.4
Id: 1/1